

Review of thermodynamics

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Equation of state

The equation of state for an ideal gas is given as,

$$p v = R T$$

or

$$p = \rho R T$$

and a calorically perfect gas is the one with constant specific heats, c_p and c_v .

Equation of state (problem)

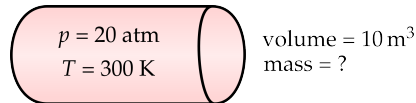
Problem 1: A pressure vessel of volume of 10 m^3 is used to store high-pressure air for operating a supersonic wind tunnel. If the air pressure and temperature inside the vessel are 20 atm and 300 K, respectively, what is the mass of air stored in the vessel?

Given: Volume = $V = 10 \text{ m}^3$,

pressure = $p = 20 \text{ atm} = 20 \times 101325 = 2026500 \text{ Pa}$,

temperature = $T = 300 \text{ K}$,

mass = $M = ?$



The density of the air in the pressure vessel can be computed using the ideal gas equation,

$$\rho = \frac{p}{R T} = \frac{20 \times 101325}{287 \times 300} = 23.54 \text{ kg/m}^3 .$$

The total mass in the pressure vessel can be computed using the equation,

$$M = \rho V = 23.54 \times 10$$

$$M = 235.4 \text{ kg} .$$

In this video...

Internal energy and enthalpy

First law of thermodynamics

Specific heat capacity

Second law of thermodynamics

Isentropic relations

Quick summary

Internal energy and enthalpy

$e \equiv$ rotational and vibrational energy of gas molecules

$$h = e + p v = e + \frac{p}{\rho}$$

The internal energy and the enthalpy have units of J/kg.

For thermally perfect gas:

$$e = e(T)$$

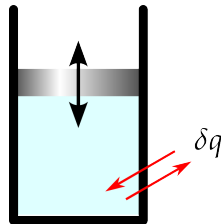
$$h = h(T)$$

First law of thermodynamics

$\delta q \equiv$ Heat added to the gas, [J/kg]

$\delta w \equiv$ Work done on the gas [J/kg]

$de \equiv$ Internal energy gained by the gas [J/kg]



The first law of thermodynamics is written as:

$$\delta q + \delta w = de$$

The work done on the system is calculated as,

$$\delta w = -p dv$$

An alternate form of the first law of thermodynamics can be written as,

$$\delta q = de + p dv$$

Specific heat capacity

The specific heat capacity is defined as $\delta q/dT$. The specific heats have unit of J/kg-K. The specific heat of gas at constant volume can be written as,

$$c_v = \frac{de}{dT} ,$$

and the specific heat of gas at constant pressure can be written as,

$$c_p = \frac{dh}{dT} .$$

For calorically perfect gases:

$$e = c_v T$$

$$h = c_p T$$

Specific heat capacity

Define ratio of specific heats : $\gamma = c_p / c_v$

From the definition of enthalpy and using equation of state for an ideal gas, we can obtain the relation,

$$c_p - c_v = R$$

Writing in terms of γ , we get:

$$c_p = \frac{\gamma R}{\gamma - 1},$$

and,

$$c_v = \frac{R}{\gamma - 1}.$$

For air, the value of $\gamma = 1.4$ and $R = 287 \text{ J/kg-K}$.

$$h = e + pv$$

$$\frac{dh}{dT} = \frac{de}{dT} + \frac{d}{dT} (pv)$$

$$\frac{dh}{dT} = \frac{de}{dT} + \frac{d}{dT} (RT)$$

$$c_p = c_v + R$$

Internal energy (problem)

Problem 2: In the previous example problem, calculate the total internal energy of the gas stored in the pressure vessel.

Given:

temperature = $T = 300 \text{ K}$,

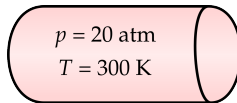
mass = $M = 235.4 \text{ kg}$ ← from problem 1.

Solution:

$$E = M e = M c_v T = M \times \frac{R}{\gamma - 1} \times T$$

$$E = 235.4 \times \frac{287}{1.4 - 1} \times 300$$

$$E = 5.067 \times 10^7 \text{ J}.$$



volume = 10 m^3
mass = 235.4 kg
internal energy = ?

Therefore, the pressure vessel has an internal energy of about 51 MJ.

Entropy and the second law of thermodynamics

Types of processes of work and heat addition to the system:

1. Adiabatic process: No heat is added to or taken away from the system.
2. Reversible process: No losses. Such as a process where effects of viscosity, thermal or mass diffusion are absent.
3. Isentropic process: Both adiabatic and reversible.

Entropy is a state variable which quantifies the deviation of a process from a reversible adiabatic process.

$s \equiv$ Entropy per unit mass

Calculation of change of entropy

The change in entropy can be calculated as,

$$ds = \frac{\delta q}{T} + ds_{\text{irrev}} .$$

where, ds_{irrev} is the increase in entropy due to friction or dissipative losses.

For a reversible process, $ds_{\text{irrev}} = 0$, otherwise $ds_{\text{irrev}} > 0$.

Therefore,

$$\boxed{ds \geq \frac{\delta q}{T}} . \quad (1)$$

If the process is adiabatic, i.e. $\delta q = 0$,

$$\boxed{ds \geq 0} . \quad (2)$$

The eqs (1) and (2) forms the second law of thermodynamics.

The second law gives us the direction of a process.

If the process is adiabatic and reversible (i.e. no losses), then,

$$ds = 0 \quad \leftarrow \text{Isentropic process}$$

Calculation of change of entropy (reversible process)

$$T ds = de + p dv$$

$$T ds = dh - v dp$$

$$s_2 - s_1 = c_p \ln \left(\frac{T_2}{T_1} \right) - R \ln \left(\frac{p_2}{p_1} \right)$$

$$s_2 - s_1 = c_v \ln \left(\frac{T_2}{T_1} \right) + R \ln \left(\frac{v_2}{v_1} \right)$$

$$(3) \quad \begin{aligned} \delta q &= de + p dv \\ \delta q &= T ds \quad \leftarrow \text{reversible process} \end{aligned}$$

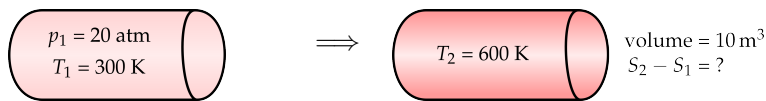
$$(4) \quad \begin{aligned} h &= e + p v \\ dh &= de + p dv + v dp \\ dh &= \delta q + v dp = T ds + v dp \end{aligned}$$

$$(5) \quad \begin{aligned} ds &= c_p \frac{dT}{T} - \frac{v dp}{T} = c_p \frac{dT}{T} - R \frac{dp}{p} \\ s_2 - s_1 &= \int_{T_1}^{T_2} c_p \frac{dT}{T} - \int_{p_1}^{p_2} R \frac{dp}{p} \end{aligned}$$

$$(6) \quad \begin{aligned} ds &= c_v \frac{dT}{T} + \frac{p dv}{T} = c_v \frac{dT}{T} + R \frac{dv}{v} \\ s_2 - s_1 &= \int_{T_1}^{T_2} c_v \frac{dT}{T} + \int_{v_1}^{v_2} R \frac{dv}{v} \end{aligned}$$

Calculation of change of entropy (problem)

Problem 3: In the pressure vessel seen in previous example, let us now heat the gas in the vessel. The heat is added until the temperature of the gas reaches 600 K. Calculate the change in entropy of the air inside the vessel.



Given: $T_1 = 300 \text{ K}$, $T_2 = 600 \text{ K}$, $M = 235.4 \text{ kg}$ ← computed in problem 1.
The volume of the vessel remains constant, $v_2/v_1 = 1$

$$s_2 - s_1 = c_v \ln \left(\frac{T_2}{T_1} \right) + R \ln \left(\frac{v_2}{v_1} \right) = \frac{R}{\gamma - 1} \ln \left(\frac{600}{300} \right) + R \ln (1)$$
$$s_2 - s_1 = \frac{287}{1.4 - 1} \ln \left(\frac{600}{300} \right)$$

Change of specific entropy: $s_2 - s_1 = 497.3 \text{ J/kg-K}$

The change of total entropy in the vessel: $S_2 - S_1 = M \times (s_2 - s_1) = 235.4 \times 497.3$

$$S_2 - S_1 = 1.171 \times 10^5 \text{ J/K}.$$

Review of thermodynamics

Isentropic relations

An isentropic process is adiabatic and reversible. Therefore, $\delta q = 0$ and $ds_{\text{irrev}} = 0$.
Hence, $s_2 - s_1 = 0$.

$$s_2 - s_1 = c_p \ln \left(\frac{T_2}{T_1} \right) - R \ln \left(\frac{p_2}{p_1} \right)$$
$$\Rightarrow \boxed{\frac{p_2}{p_1} = \left(\frac{T_2}{T_1} \right)^{\gamma/(\gamma-1)}}$$

$$s_2 - s_1 = c_v \ln \left(\frac{T_2}{T_1} \right) + R \ln \left(\frac{v_2}{v_1} \right)$$
$$\Rightarrow \boxed{\frac{v_1}{v_2} = \frac{p_2}{p_1} = \left(\frac{T_2}{T_1} \right)^{1/(\gamma-1)}}$$

$$0 = c_p \ln \left(\frac{T_2}{T_1} \right) - R \ln \left(\frac{p_2}{p_1} \right)$$
$$R \ln \left(\frac{p_2}{p_1} \right) = c_p \ln \left(\frac{T_2}{T_1} \right)$$
$$R \ln \left(\frac{p_2}{p_1} \right) = \frac{\gamma R}{\gamma-1} \ln \left(\frac{T_2}{T_1} \right)$$

$$0 = c_v \ln \left(\frac{T_2}{T_1} \right) + R \ln \left(\frac{v_2}{v_1} \right)$$
$$0 = \frac{R}{\gamma-1} \ln \left(\frac{T_2}{T_1} \right) + R \ln \left(\frac{v_2}{v_1} \right)$$
$$R \ln \left(\frac{v_1}{v_2} \right) = \frac{R}{\gamma-1} \ln \left(\frac{T_2}{T_1} \right)$$

Isentropic relations

The isentropic relations can be combined into a single expression,

$$\frac{p_2}{p_1} = \left(\frac{\rho_2}{\rho_1} \right)^\gamma = \left(\frac{T_2}{T_1} \right)^{\gamma/(\gamma-1)}$$

Summary of important relations

For perfect gas:

$$c_p = \frac{\gamma R}{\gamma - 1} \quad ; \quad c_v = \frac{R}{\gamma - 1}$$

where:

$$R = \frac{R_u}{\hat{m}} \quad ; \quad R = c_p - c_v \quad ; \quad \gamma = \frac{c_p}{c_v}$$

The first law of thermodynamics is given by:

$$\delta q + \delta w = de$$

Specific enthalpy is defined as:

$$h = e + p v = e + \frac{p}{\rho}$$

Summary of important relations

The second law of thermodynamics gives the relation for change of entropy:

$$ds = \frac{\delta q}{T} + ds_{\text{irrev}} \quad ; \quad ds \geq \frac{\delta q}{T}$$

for reversible process:

$$ds = \frac{\delta q}{T}$$

for reversible-adiabatic process (isentropic):

$$ds = 0$$

Relations linking first and second law of thermodynamics:

$$T ds = de + p dv$$

$$T ds = dh - v dp$$

Summary of important relations

Change in entropy for calorically perfect gas:

$$s_2 - s_1 = c_p \ln \left(\frac{T_2}{T_1} \right) - R \ln \left(\frac{p_2}{p_1} \right)$$

$$s_2 - s_1 = c_v \ln \left(\frac{T_2}{T_1} \right) + R \ln \left(\frac{v_2}{v_1} \right)$$

Isentropic relation:

$$\frac{p_2}{p_1} = \left(\frac{\rho_2}{\rho_1} \right)^\gamma = \left(\frac{T_2}{T_1} \right)^{\gamma/(\gamma-1)}$$